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Germline variant analysis of family trios in children with cancer

Background

Childhood cancers are rare diseases, for which a pathogenic germline variant can be identified in about 10% of all patients. The analysis of trio sequencing data from families consisting of parents and a cancer-affected child can thus shed light onto cancer-associated germline variants, their inheritance patterns, and the role of digenic and multigenic interactions in tumorigenesis. The Düsseldorf germline study was started in 2015 and focuses on the analysis of parent-child trio whole-exome sequencing (WES) data. More than 200 parent-child WES trios have been generated in the context of this study; an overview of the cancer diagnoses of the first 160 trios is provided in Figure 1.

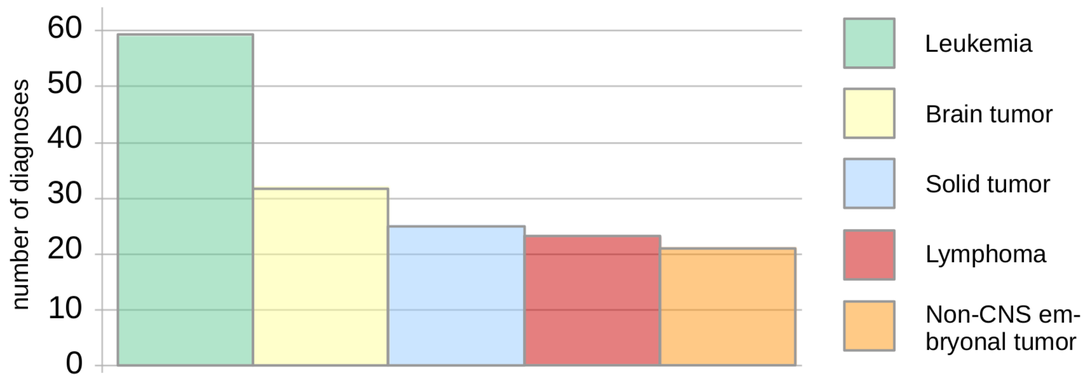

Figure 1: Distribution of the first 160 germline trios' cancer diagnoses. Leukemia was the largest cohort with 59 cases, followed by brain tumors.

Methods

We use a pipeline approach based on a combination of three variant calling algorithms to identify transmitted and de novo germline variants in whole exome sequencing (WES) data from parents-child trio samples. The mean on-target coverage per sample is 80x or higher for the majority of the trios; samples with lower on-target coverage are resequenced.

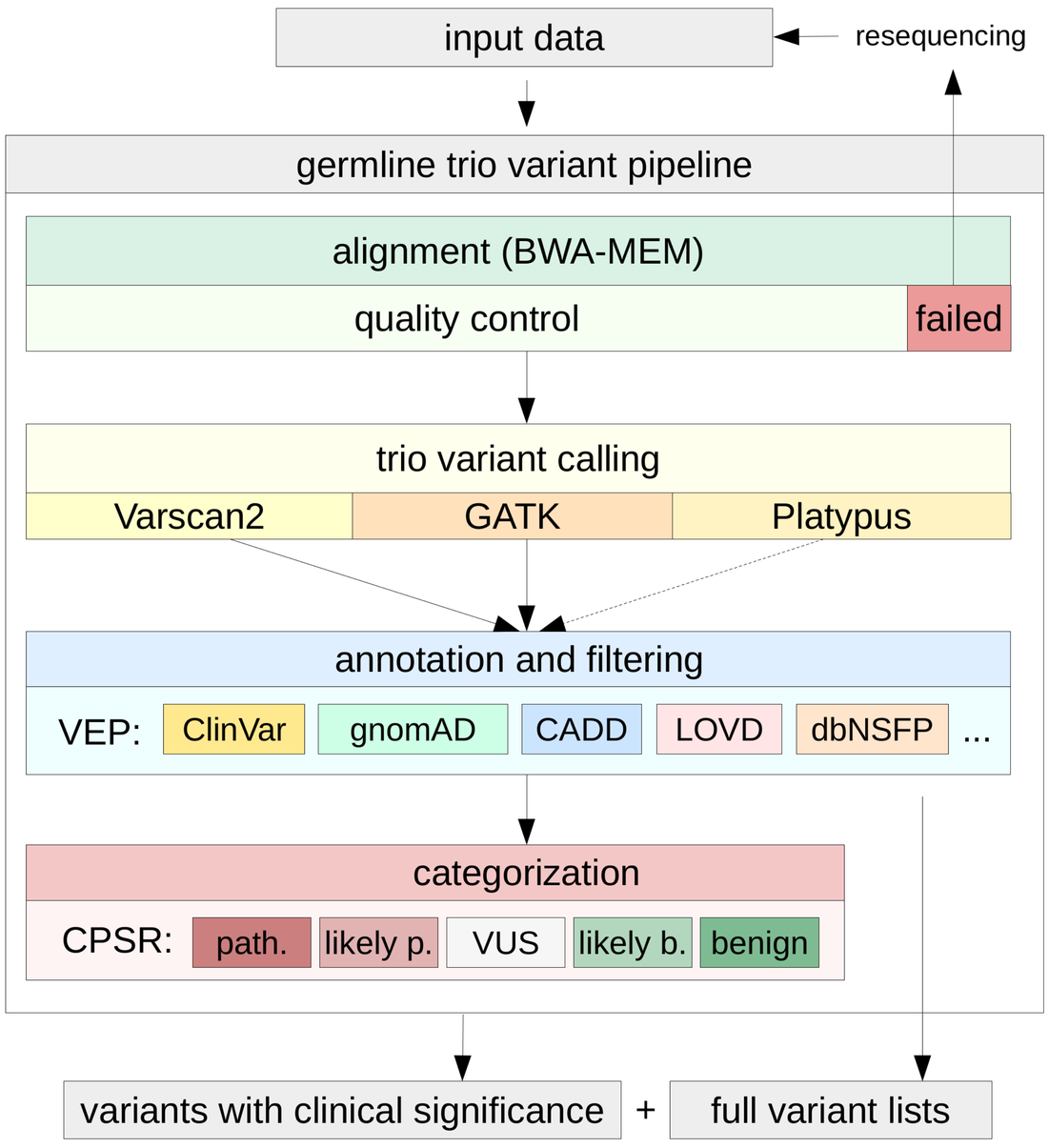

Figure 2: Pipeline workflow: After germline variant calling with three algorithms, the resulting variants are annotated with Ensembl's VEP tool. Variants in target genes are classified with the algorithm CPSR as (likely) pathogenic, variants with uncertain significance (VUS), or (likely) benign.

After a standardized alignment and quality control step with FastQC [1] and qualimap [2], single nucleotide variants (SNVs) and insertions / deletions (indels) are identified for each family trio and algorithm, and integrated with a combination of union and overlap strategies. A union of Varscan2's trio mode [3] and GATK's trio approach [4] is used for SNV calling, while a union of GATK's variants with the overlap of Varscan2's and Platypus's [5] results is used for the identification of indels. The resulting variant list per sample is subsequently annotated with Ensembl's VEP algorithm [6] and a variety of prediction scores and state-of-the-art database information, and subsequently filtered to remove known polymorphisms. Comparison against an in-house cohort of 158 trios allows for rigorous filtering of technical artifacts. Output options include full overview tables for the complete trio variant results, filtered lists for specified gene sets, and an integrated report from the Cancer Predisposition Sequencing Reporter (CPSR, [7]), a pre-published algorithm that performs automated variant interpretation in known cancer predisposition genes. The pipeline's workflow is shown in Figure 2. The whole pipeline is embedded into a Docker container with a Conda environment, and configured to run semi-automatically for a specified set of WES trio data.

Results and discussion

We called approximately 500.000 raw variants per trio, of which 600-900 filtered variants were included in the CPSR target gene analysis. An example output of the program's summary plots is given in Figure 3, CPSR's full output contains further information regarding the variants' symbol, ClinVar phenotype, their consequence, and potential protein changes (if applicable). The germline trio pipeline's main output consists of prefiltered variants in .csv or text format, with detailed annotations from Ensembl's VEP tool and databases like ClinVar [8], the gnomAD non-cancer cohort [9] or CADD [10]. Hence, the trio pipeline facilitates concentration on the most relevant germline alterations in the context of cancer predisposition, as well as the analysis of the broader mutation landscape in childhood cancer.

A preliminary set of variants in well-known cancer genes like *TP53*, *MSH6* and *DICER1* was validated through Sanger sequencing. Still, further validation experiments and test runs on simulated synthetic data are required to thoroughly investigate the current pipeline setup's performance with regards to its precision and recall.

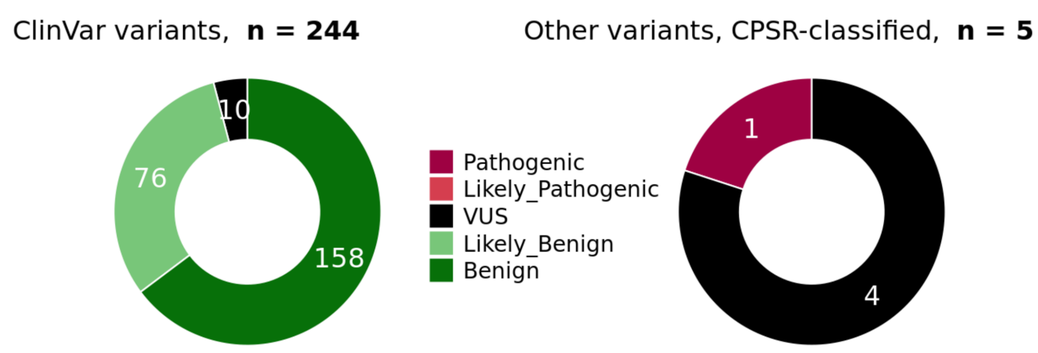

Figure 3: Example output for CPSR's classification of variants. Variants with a ClinVar annotation (left) and other variants with a CPSR classification (right) are presented separately.

References

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